



MA 001 Preparatory Mathematics - 1

Course Code: MA 001

Course Name: Preparatory Mathematics - 1

L-T-P-C: 3-1-0-4

Students Intended for: Preparatory Students

Core or Elective: Core

Approval: 50th BoA

Course Contents:

- **Basic Set Theory:** Functions and their inverses, composition of functions, relations, equivalence relations, partitions [6 Lectures]
- **Number Systems:** Numbers – natural numbers, integers, rationals, reals, complex, uncountability of reals, irrationality of 2 etc. Congruencies, Residue Classes, addition and multiplication modulo n etc. [8 Lectures]
- **Complex Numbers:** Complex Numbers as ordered pairs. Argand's diagram. Triangle inequality. De Moivre's Theorem. [4 Lectures]
- **Algebra:** Quadratic equations and expressions; permutations and combinations; Binomial theorem for positive integral index [4 Lectures]
- **Coordinate Geometry:** Locus, Straight lines; Equations of circle, parabola, ellipse and hyperbola in standard forms; parametric representation [8 Lectures]
- **Vectors:** Addition of vectors. Multiplication by a scalar; scalar product, cross product and scalar triple product with geometrical applications. [6 Lectures]
- **Matrices and Determinants:** Algebra of matrices; Determinants and their properties; Inverse of a matrix; Cramer's rule. [6 Lectures]

Text Books:

1. S. Lang, **An Introduction to Linear Algebra**, undergraduate text in Mathematics, Springer Verlag.
2. Fred Safier, **Schaum's outline of Precalculus**, 2nd Edition, McGraw Hill, 2008.
3. S. L. Loney, **The elements of Coordinate Geometry**, Scholarly publishing office, University of Michigan Library, 2005.

Reference Books:

1. Murray R. Spiegel, **Schaum's Outlines of Vector Analysis (and An Introduction to Tensor Analysis)**, McGraw Hill, 1968.
2. Murray R. Spiegel, **Schaum's Outlines: Complex Variables (with An Introduction Conformal Mapping and its Applications)**, McGraw Hill, 1964.
3. R. K. Jain and S. R. K. Iyengar, **Advanced Engineering Mathematics**, 3rd Edition, Narosa Publisher.